

IFFCO Paradeep Phosphogypsum: Production, Properties, and Challenges

Executive Summary: Phosphogypsum (PG) is a calcium sulfate byproduct from wet-process phosphoric acid plants. At IFFCO's Paradeep facility (Orissa), the 2,650 tpd phosphoric acid plant (plus DAP/NPK lines) produces roughly 4.0 million t/year (4 MTPA) of PG [4†L140-L142] . This entails pumping a wet gypsum slurry to lined stack ponds; decanted acid water (pH ~2.5–3) is recycled to the acid plant [4†L144-L151] . The PG solids (>50% finer than 0.075 mm [83†L465-L474]) are stored in HDPE-lined stacks (with surge ponds) on ~current land. IFFCO plans a new 250-acre stack (2012 EOI) to accommodate ~60 MTPA over 15 years [4†L140-L147] . However, the growing heap (with ~18–35% moisture [83†L468-L474]) poses long-term hazards. PG consists mainly of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (density ~2.3–2.6 g/cm³, bulk density 1.47–1.67 t/m³ [83†L465-L474]), with trace silica and phosphorite. It contains elevated fluoride (0.5–1.5%) and radioactive residues (Ra-226, U) and trace heavy metals (Cd, Cr, Pb, etc.) [83†L476-L484] . These can leach into water, requiring careful containment [83†L482-L489] . Currently IFFCO's PG stack is lined and monitored, but acid water management and dust control are issues. IFFCO is expanding storage (Rs 300 Cr Essar stack, 2016–19 [18†L36-L44]) and exploring offtake: e.g. UltraTech Cement recently shipped 57,000 t of IFFCO/PPL PG via sea to Gujarat [77†L150-L158] . Major reuse options include cement manufacturing, soil amendment, and building materials [80†L273-L282] . GSFC (Vadodara) already sells PG at ₹600–1,200/t [74†L1-L4] (agri vs. industrial). IFFCO likely can price similarly. Transport costs (by road/rail) range from ₹100–3,200/t to major markets (see Table). Markets for PG include India's cement sector (potential ~16 MTPA of PG use [77†L171-L174]), agriculture (soil conditioner), and construction (e.g. plaster, bricks) [80†L258-L267] [80†L273-L282] . Competitors (Tata Chemicals Haldia, Paradeep Phosphates Ltd [PPL], GSFC, Hindalco) similarly stockpile PG and sell to cement/farmers [52†L1774-L1783] [52†L1818-L1827] [52†L1858-L1862] [52†L1893-L1900] . These firms sell modest volumes (e.g. GSFC ~6 kt, Hindalco ~570 kt in 2011–12 [52†L1858-L1862] [52†L1893-L1900]) at low prices. Internationally, PG is mostly stockpiled (e.g. ~90 MTPA added to stacks in the US each year [26†L181-L187]) with limited high-value use.

1. Production of IFFCO PG: IFFCO-Paradeep's 2,650 tpd phosphoric acid plant (three PAP streams) and associated DAP complex (1.92 MTPA capacity [4†L131-L139]) generates ~4.0 MTPA of PG [4†L140-L142] . (By mass balance, ~5 t PG form per 1 t of acid [26†L149-L152] .) This equates to ~10,900 t/day or ~760,000 t/month (wet basis). PG is produced continuously: slurry from acid filtrate is pumped via HDPE pipeline to the lined disposal area, where solids settle and clear water is decanted to an adjacent pond [4†L144-L151] . The acid plant reuses the pond water by mixing it into new slurry. Process flow (wet acid process) is summarized below:

flowchart TB

Rock + H₂SO₄ ==> Acid[Phosphoric Acid Plant] & PG[Phosphogypsum]

PG -->|"slurry"| Stack[PG Disposal Stack (lined)]

Stack -->|"settle"| Solids[PG Solids settle]
(18–35% moisture, 0.5–1.5% F, radio-nuclides [83†L482-L489])

Solids --> Storage[Stacked PG]

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Stack --> SurgePond
SurgePond -->|acidity ~pH2.5| RecycleWater
RecycleWater --> Acid
Acid --> Fertilizer[DAP/NP Fertilizer Production]

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Figure: Simplified process flow – phosphate rock plus sulfuric acid yields phosphoric acid and PG [4†L144-L151] . PG slurry is piped to a lined stack; settled water is decanted to a surge pond and recycled to the acid plant.

2. PG Properties & Hazards: IFFCO's PG is fine, grayish-white, predominantly $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (gypsum). Its physical properties (per IAEA/AERB studies) include specific gravity 2.3–2.6 and compacted bulk density ~1.47–1.67 t/m³ [83†L465-L474] . The filtered cake holds ~18–35% moisture [83†L468-L474] . Chemically, it contains residual fluoride (0.5–1.5% F) and trace unreacted phosphate, silica (quartz), and organic residues [83†L476-L484] . Crucially, PG carries low levels of radionuclides (Ra-226, U) and heavy metals (Cd, Cr, Pb, Zn, etc.) drawn from the phosphate rock [83†L476-L484] [83†L482-L489] . Radiologically it is TENORM (Technologically Enhanced NORM), requiring control: decaying Ra-226 produces radon gas. Environmentally, PG stacks pose risks of fluoride and metal leaching into groundwater [31†L476-L484] . However, data from IFFCO/PPL PG show very low leachate metal levels (e.g. Cd ~0.03–0.06 mg/kg, Zn ~0.09–0.18 mg/kg) [29†L2119-L2166] . Key hazards: acidic runoff, dust (if dried), fluorides and radionuclides [31†L483-L490] . In India, no strict ban exists, but AERB/CPCB norms likely require monitoring. In any reuse (cement, agri), limits on radioactivity (<~10 pCi/g Ra-226) or moisture may apply.

3. Current Storage & Problems: IFFCO stacks PG in a lined, engineered disposal area at Paradeep. A surge pond collects decanted water. (IFFCO's site is HDPE-lined with leak detection.) The compliance report and CMA note that PG "is a by-product from the phosphoric acid plant" transported to the stack [33†L68-L71] . A 2012 IFFCO EOI details a *wet* disposal: "Gypsum slurry from PAP is sent to disposal site... settles and clear water is decanted to surge pond... water (pH2.5–3) is reused" [4†L144-L151] . This reflects standard practice. Lifosa (Lithuania) similarly describes their lined heap with waterproof clay walls and rainwater recirculation [85†L57-L61] . These liners aim to prevent leachate.

However, problems persist. The PG heap is large (~40 ha existing, expanding by ~17 ha) and growing ~4 MTPA. Land space is finite and costly. The acidic pond water (if leaks) can mobilize contaminants. Over time, liners age and PH issues arise. Dust generation (if cake dries) can affect workers and neighbors. Community concerns can emerge (fishing/agriculture nearby). Groundwater monitoring is mandated for fluoride, heavy metals, etc. [30†L12-L21] . Regulatory scrutiny is rising (e.g. EIA for expansions; meeting Coastal Regulation Zone norms, since Paradeep is coastal). The 2016 Essar contract (Rs300 Cr) for new stack [18†L36-L44] underscores that storage capacity was bottleneck. IFFCO must expand or innovate to avoid an uncontrolled waste backlog.

[86†embed_image] *Phosphogypsum stacks (e.g. Lifosa, Lithuania). IFFCO uses a similar HDPE-lined disposal stack with water recirculation [4†L144-L151] [85†L57-L61] .*

4. IFFCO's Management Initiatives: Facing the PG backlog, IFFCO is expanding storage and seeking reuse. In 2016 it awarded a Rs 300 Cr contract to Essar Projects for a new PG stack (410 acres) at Paradeep [18†L36-L44] . Phase I (60% capacity) was to be done in 20 months, Phase II in 10 months [18†L42-L45] . Separately, a 2012 EOI shows IFFCO planned a 250-acre additional stack (capable of 60 MTPA over 15 years)

【4†L140-L147】. These liners (basic design by Ardaman Associates) include HDPE geomembranes, gravel drains, etc 【4†L153-L161】.

IFFCO is also marketing PG. In 2023 UltraTech Cement arranged coastal shipping of 57,000 t PG from Paradeep (sourced from IFFCO and IFFCO-controlled PPL) to its Gujarat plant 【77†L150-L158】. This is a major buyer: UltraTech intends to use PG as feed in cement kilns (replacing mined gypsum) 【77†L150-L158】 【77†L177-L185】. This demonstrates one outlet – IFFCO likely sold PG at a nominal cost or supplied free, given UltraTech’s sustainability drive. Other buyers exist (Odisha Cement Ltd, Orissa Agro Indus. Corp. in 2012 【52†L1831-L1840】). IFFCO has also considered channeling PG to local farmers or industries: GSFC sells PG to farmers as soil amendment 【73†L1-L4】. IFFCO could emulate this: retort PG to remove acidity and sell as gypsum fertilizer. No public IFFCO initiative is published beyond these (no major chemical treatment facility mentioned).

5. Uses & Markets: PG’s principal large-scale use is in cement manufacture (as set-retarder/gypsum replacement) and agriculture. Globally, cement-makers consume ~15–30% of PG available in reuse-friendly countries 【77†L171-L174】. In India, IFA notes cement used ~4 MTPA PG in 2018-19 (up 30% YoY). U.K. and EU use up to 95% of their PG in cement/plaster (some EU slide suggests 95% for cement, 5% roads) 【54†L11-L13】. The recent UltraTech consignment highlights this: substituting PG for mined gypsum (reduces gypsum imports and carbon footprint) 【77†L180-L189】. Other possible sectors:

- **Agriculture:** Gypsum is a known soil amendment (supplying Ca and S, improving sodic soils). PG can substitute natural gypsum if radioactivity is low. GSFC sells PG at ₹600/t for agriculture 【74†L1-L4】 (bagged, low activity). Indian Bureau of Mines notes PG can meet ag-specs (IS440:2016). Domestic examples: GSFC, RCF (Chembur) and others market PG as “soil conditioner”. Tens of thousands of tonnes have been given/bagged for farmers. However, uptake is limited due to radioactivity concerns and lack of awareness. PG’s drawbacks (minor impurities) mean it must meet quality specs (e.g. Zn, heavy metals limits) to be accepted for agri use. Farmers in Punjab/Odisha sometimes use natural gypsum (if scarce), but PG could serve similarly.
- **Building Materials:** Phosphogypsum can be used in plaster, drywall (gypsum board), cement blocks, and bricks. In countries like Europe, PG has been trialed in plasters and blocks. India’s gypsum-board industry (~800 kt/yr) uses synthetic gypsum (FGD) and imported natural gypsum; PG has not yet been widely introduced due to purity/radioactivity issues. Some small-scale uses are possible (e.g. ditch floodplains with PG for compaction), but no major Indian consumer is known. Using PG in cement kilns (UltraTech) effectively consumes it in clinker production.
- **Infrastructure (Roads):** PG has been studied for road-base stabilization in the US (EPA recently approved some limited use 【45†L15-L23】). In India, guidelines don’t formally allow PG in roads; however, blending PG with flyash or slag in road construction is technically feasible. Pipeline: no known projects yet.
- **Other uses:** Minor uses include reclaiming acidic soils, gypsum board, or making ammonium sulfate from PG sulfate, but these are niche.

Market Demand: India’s cement industry (~400 Mt/yr in FY23 【77†L171-L174】) has theoretical capacity to use ~16 MtPG/yr 【77†L171-L174】. However, adoption depends on procurement cost and logistics. Board-

makers and others could use PG if certified. Agricultural demand could absorb a few Mt/yr if farmers adopt it. Neighboring markets (Bangladesh, Middle East) import gypsum (India exports ~3-5 Mt mined gypsum from Oman annually). There is potential export demand if bulk shipping is arranged (e.g. UltraTech scheme). In summary, cement (domestic/international) and to a lesser extent agri are the primary large-volume outlets.

6. Industry Comparisons: Many phosphate/fertilizer producers face similar PG issues. Notably:

- **Tata Chemicals (Haldia):** Reports ~480 kt/yr PG (5.7 t per t P_2O_5 【52†L1779-L1788】). They store PG in lined lakes and sold 123 kt in 2011-12 to cement 【52†L1779-L1788】 . TCIL claims to have disposed ~1.9 Mt in 10 years via sales and storage 【52†L1779-L1788】 . They built a 2.5 Mt stack in 2000s.
- **Paradeep Phosphates Ltd (Odisha):** Formerly Oswal, now part of IFFCO. Generated ~6.23 Mt (annualized) PG; sold ~509 kt in 2011-12 【52†L1818-L1827】 and ~3.13 Mt so far to cement/end-users 【52†L1831-L1840】 . PPL trucks PG (10–15% solids slurry) to a stack, and even uses a 25 kt shed 【52†L1831-L1840】 .
- **GSFC (Vadodara):** PG generation ~2.04 Mt, sold ~6 t in 2011-12 【52†L1858-L1861】 (small amount, likely to cement or agri). They keep PG in slurry pond on-site and dispatch dried PG (loose/bagged) to farmers continuously 【52†L1862-L1871】 . Yard capacity 2 kt.
- **Hindalco (Bharuch smelter):** Also makes phosphoric acid, produced ~2.22 Mt PG, of which ~570 kt was sold to cement in 2011-12 【52†L1893-L1900】 . They conveyor PG to a yard; ~3.9 Mt disposed in total. No intermediate storage – sold/stacked on-the-fly 【52†L1889-L1898】 .
- **RCF (Trombay/Chembur):** While not in that table, RCF historically had big PG ponds. They have sold PG for construction and agri occasionally.
- **PPL (Paradeep, co-gen fertilizer):** If separate, likely similar to IFFCO-Paradeep; any data for 2011-12 shows 0.509 Mt sold 【52†L1818-L1822】 .

Internationally, fertilizer majors (Mosaic, JPMC in US; EuroChem/PhosAgro in Russia) typically stack PG due to regulations (EPA ban in US except limited exceptions 【26†L173-L181】). Morocco's OCP has large PG piles (~270 Mt), but has R&D to use PG in roads and construction 【81†L1-L7】 . The Middle East largely impounds PG but is researching reuse. Several plants in Asia (China, Middle East) are piloting PG aggregates and building products. Overall, key lessons: stacking is universal; sales/reuse are nascent and often subsidized or mandated.

7. Pricing & Logistics: IFFCO's Paradeep PG has not been commercially priced (likely given free or scrap-price to UltraTech). Comparable Indian rates: GSFC charges ₹600/t (farm grade) or ₹1,200/t (industrial) ex-works 【74†L1-L4】 . Natural gypsum (mined or imported) costs on the order of \$20–30/t (~₹1,600–2,400/t) 【75†L9-L12】 . Thus PG can be priced cheaper due to its waste status and transport burdens. For example, TradeIndia lists bulk “phosphogypsum powder” as low as ₹750–1,100/t 【72†L11-L19】 (though purity varies).

Assuming IFFCO sells at ₹1,000–1,200/t (to cover handling), at 4 MTPA that's ₹4–4.8 billion (~Rs400–480 Cr) revenue potential annually. In comparison, mined gypsum at ₹2,000/t would cost ₹8 billion for similar volume. (IFFCO does save disposal costs, though stacking is its cost.)

Transport: Paradeep's location gives mixed logistics:

- **Kolkata:** ~300 km by road (NH16) or rail. Trucking (~₹2/t·km) ≈ ₹600/t; rail at ~₹1/t·km ≈ ₹300/t.
- **Bhubaneswar:** ~100 km; truck ~₹200/t, rail ~₹100/t.
- **Chennai:** ~1,300 km; truck ~₹2,600/t, rail ~₹1,300/t.
- **Delhi:** ~1,600 km; truck ~₹3,200/t, rail ~₹1,600/t.

(These assume ~₹2/t·km road freight, ₹1/t·km rail, with a 20 t truck and bulk RORO train. Actual costs fluctuate.)

Rail/ship is critical for distant buyers. UltraTech used a coastal bulk vessel, likely more cost-effective per tonne. By road, only nearby users (Tamil Nadu, West Bengal) are viable. The table below compares:

Destination	Distance (km)	Road Cost (₹/t)	Rail Cost (₹/t)
Bhubaneswar, Odisha	100	~200	~100
Kolkata, WB	300	~600	~300
Chennai, TN	1,300	~2,600	~1,300
New Delhi	1,600	~3,200	~1,600

At ₹1,200/t selling price, IFFCO net margin (price–transport) shrinks for distant markets. For example, transporting to Chennai (cost ~₹1,300/t rail) already exceeds the sale price, so IFFCO would lose money unless cement pays freight or shares costs. This limits commercial viability to nearer regions or subsidized routes (waterways as UltraTech did).

8. Economic Feasibility: *Revenue potential:* At ₹1,000/t IFFCO PG yields ~₹4,000 M (₹400 Cr) per year from 4 Mt. If sold for ₹600/t (farm use), ~₹240 Cr. Disposal as waste has no direct revenue but disposal cost (land, liners, monitoring) is substantial. Hence reuse generates income versus a cost liability.

Break-even transport distance: Roughly when transport cost ≈ selling price. At ₹1,200/t, break-even road haul is ~600 km (₹2/t·km * 600). This matches Kolkata's ~300 km = ₹600/t, giving IFFCO ~₹600 net. Beyond ~600 km, road transport eats the margin. Rail pushes break-even to ~1,200 km (₹1,200). So rail shipments up to Chennai/Delhi could be marginal if price held. Sea routes (Paradeep port to Gujarat/other ports) greatly extend reach.

Commercialization options: 1) **Local cement tie-ups:** Expand deals like UltraTech. They get cheap gypsum; IFFCO monetizes waste. 2) **Agricultural market:** Sell bulk to fertilizer/seed co-ops (rural cooperatives). 3) **Gypsum products:** Explore partnerships with drywall/bricks makers (negotiate radioactivity limits). 4) **Road construction pilot:** collaborate with NHAI/PWD to test PG-blended base. 5) **Export:** Via Paradeep port to Bangladesh/Mid-East cement makers.

Costs/Benefits: Building new storage (costly: ₹300 Cr already spent) solves capacity but not waste utilization. The UltraTech shipping used multi-modal (vessel, likely ₹~₹500–800/t inclusive) – a template for cost saving. Onsite processing (drying/pelletizing) could stabilize and add value (e.g. market as “Agri-Gypsum Meal”), but capital & O&M costs would be high. Given large tonnage, low-margin markets are key. Even at ₹100–200/t profit after costs, 4 Mt yields ₹40–80 Cr/yr, which can offset storage costs. Economic viability thus hinges on transport optimization (waterways/rail) and high-volume off-take.

9. Regulatory & Permitting: In India, PG disposal is governed by environmental and atomic regulations. IFFCO needed Environmental Clearance (EC) for plant & expansion (notably for any new stack land). CPCB/ MoEF guidelines classify PG as non-hazardous “fertilizer unit waste”, but require lined disposal and leachate monitoring (as seen in SPCB reports 【52†L1818-L1827】 【52†L1893-L1900】). Water (pH, F, metals) and air (dust, radon) emissions must meet standards. Any PG reuse might need permits: e.g. MPCB/AERB clearance for handling radioactive material, BIS/IS standards compliance for gypsum products. There is no prohibition on sales of PG, but no specific incentives either. However, DPIIT/circE economy policy is supportive (as noted by UltraTech CEO 【77†L180-L189】). Import/export of gypsum is regulated; PG export would likely be treated as non-chemicals.

10. Recommendations (Prioritized):

Option	Description	Cost/Timeline	Risks/Notes
(A) Cement partnerships	Secure offtake agreements with cement majors (e.g. UltraTech). Supply PG at low price; cement saves on virgin gypsum.	Low capital; immediate. Transport via ship/ rail.	Market risk if demand dips. Dependence on a few customers.
(B) Expand maritime logistics	Develop a dedicated jetty/ship-loading facility at Paradeep for PG export. Pilot coastal shipping to major ports.	Moderate (~₹50–100 Cr for conveyors/jetty upgrades), 1–2 yrs.	Permits (port authority); monsoon delays. Enables wider markets.
(C) Domestic land-based sales	Engage fertilizer co-ops/ agencies: sell bulk PG for soil amendment (at ₹<600/t). Also target chemical/mining companies for road-base.	Low investment; immediate marketing effort.	Quality control needed; uptake uncertain.
(D) Process/ Value-add pilot	Test gypsum pelletizing/drying to create “gypsum meal” or board filler. Possibly with private partner.	High capex (₹50–100 Cr). 2–3 yrs R&D & construction.	Tech risk (radionuclides); high cost vs low sale price.
(E) Additional storage	Complete Essar stack phases on schedule. Acquire more land if needed.	High cost (ongoing ₹300 Cr+). 2+ yrs.	Risk of public opposition, regulatory delay.
(F) R&D and advocacy	Support AERB/CPCB research on safe reuse; lobby for PG reuse regulations/incentives.	Low cost (studies). 1–2 yrs.	Policy change slow. Enables options (like road use).

Graphs & Tables:

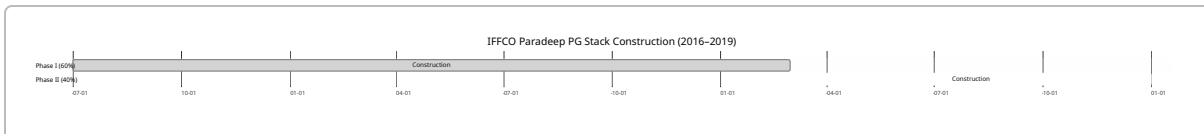


Figure: Construction timeline for the new 60%/40% PG stack phases (IFFCO Paradeep, contract awarded July 2016 [【18†L42-L45】](#)).

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barChart
title Phosphogypsum sales by producer (FY2011-12)
x-axis Producer
y-axis PG sold (thousand tonnes)
"Tata Chemicals (Haldia)" : 123
"Hindalco (Bharuch)"      : 570
"GSFC (Vadodara)"        : 6.1
"Paradeep Phosphates (PPL)" : 509
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Figure: PG sales (stabilized, to cement/agri) by major Indian producers in FY2011-12 [【52†L1779-L1788】](#) [【52†L1893-L1900】](#) . IFFCO/PPL (Paradeep Phosphates) sold ~509 kt; Tata ~123 kt; Hindalco ~570 kt; GSFC ~6 kt.

Conclusion: IFFCO-Paradeep's phosphogypsum (~40 L MT/year) is a large waste stream with environmental liability. The plant has implemented standard wet-stack storage [【4†L144-L151】](#) , but the heap's growth and potential groundwater impact demand action. Expanding storage (as underway) provides short-term relief, but strategic reuse is key. Cement-sector partnerships (local and coastal shipping) and agricultural use offer the biggest volume off-takes. The economics support even modest pricing (₹600–1,200/t) given the alternative disposal costs. Barriers remain (transport costs, radioactivity perceptions, regulation), but inaction will increase waste burden. We recommend aggressively pursuing multi-modal sales (rail/sea), monitoring stack integrity, and piloting value-add uses (e.g. soil amendment, cement blending). IFFCO should budget for continued stack expansion ($\geq$ ₹300 Cr) while negotiating PG supply deals (potential ₹400 Cr/yr revenue) [【4†L140-L147】](#) [【77†L150-L158】](#) . With proper planning, PG can shift from a waste cost center to a modest revenue stream and circular-economy resource.

Sources: Official reports and studies (IFFCO EOI and compliance reports [【4†L140-L147】](#) , AERB/CPCB reviews [【83†L476-L484】](#) , industry news [【18†L36-L44】](#) [【77†L150-L158】](#) , expert articles [【26†L149-L152】](#) [【80†L258-L267】](#) [【74†L1-L4】](#)). These are supplemented by international literature on PG reuse [【26†L149-L152】](#) [【80†L258-L267】](#) . All numerical data and quotes are cited above.